

VIP Engineering Review Report

Team D: The Last Set

Autonomous Biomechanical Kinematics and AI-Driven Form Correction System for Strength Training

Faculty Technical Review, Research Roadmap, and 50-Point Evaluation Rubric

IMU Sensor Fusion → EKF Kinematic Modeling → Vision-Based Pose Estimation → Dynamic Time Warping Analysis

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Course	Vertically Integrated Project, VIP Course 2026–1
Review Target	Team D: The Last Set
Total Evaluation Scale	50 points
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Executive Summary

This report comprehensively evaluates the engineering proposal submitted by VIP Team “The Last Set.” The project targets a critical gap in the digital fitness and physical therapy markets by engineering a multimodal biomechanical tracking system. By fusing 9-axis Inertial Measurement Unit telemetry with edge-based optical pose estimation, the system mathematically evaluates resistance-training kinematics and provides real-time injury-prevention alerts and structural form correction. The evaluation scrutinizes the mathematical rigor of the sensor-fusion algorithms, the hardware architecture, the real-time feedback pipeline, and the team’s long-term commercialization strategy.

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1 Feedback Report

Executive Summary

This technical research evaluation rigorously reviews the Vertically Integrated Project (VIP) submitted by Team “The Last Set.” The contemporary digital fitness industry, spanning commercial smartwatches, home-gym equipment, connected coaching platforms, and physical therapy rehabilitation trackers, is fundamentally constrained by a lack of reliable 3D spatial awareness. Current commercial devices rely heavily on 3-axis accelerometers, gyroscopes, and photoplethysmography (PPG) sensors to measure localized activity and heart rate. However, these systems are largely incapable of analyzing complex kinematic chains, joint angles, limb-segment coordination, or muscular load distribution during resistance training.

Consequently, athletes, elderly users, rehabilitation patients, and general fitness consumers remain vulnerable to structural injuries caused by improper biomechanical execution during movements such as squats, deadlifts, bench presses, lunges, rows, and overhead presses. The key technical challenge is not merely counting repetitions; it is identifying whether the repetition is mechanically safe, symmetrical, stable, and consistent with a user-specific movement profile.

Team “The Last Set” addresses this systemic inefficiency by architecting a hybridized biomechanical tracking pipeline. The proposed system integrates wearable 9-axis Inertial Measurement Units (IMUs) equipped with localized microcontrollers, such as ESP32 boards, alongside an external computer vision module using MediaPipe or OpenPose for full-body spatial localization. By combining the high-frequency temporal resolution of IMUs with the absolute spatial referencing of optical tracking, the system mitigates the classical quaternion drift problem inherent in purely inertial systems.

The fundamental design rationale centers on real-time edge processing. Rather than transmitting raw tensor data, video streams, and high-frequency IMU telemetry to cloud servers, which would introduce unacceptable latency for real-time form correction, the team localizes Extended Kalman Filter (EKF) and Dynamic Time Warping (DTW) algorithms on edge nodes. The resulting kinematic profiles are compared against mathematically modeled “golden repetitions” derived from professional biomechanical data.

The anticipated long-term outcomes for this VIP trajectory extend beyond a single-semester deliverable. By establishing a rigorous, documented pipeline for data acquisition, sensor fusion, structural processing, kinematic scoring, and real-time auditory or haptic feedback, the current cohort provides an extensible foundation. Subsequent cohorts are expected to inherit this framework and integrate advanced methodologies, including individual-specific anthropometric scaling, predictive injury modeling using Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Temporal Convolutional Networks (TCNs), and seamless integration into clinical rehabilitation protocols.

Business and Market Insight Why This Project Matters

The intersection of biomechanics, wearable technology, Edge AI, and personalized digital coaching represents a multi-billion-dollar market spanning consumer fitness, sports performance, workplace ergonomics, and clinical physical therapy. A low-cost, high-fidelity kinematic tracker gives students a practical research platform where embedded systems, sensor mathematics, digital signal processing, computer vision, and applied machine learning converge into a highly marketable engineering workflow.

2 Project Metadata and Technical Scope

Table 1: Project Metadata and Review Scope

Item	Description
Project Name	Team: The Last Set – AI-Driven Form Correction System
Primary Objective	Develop an end-to-end multimodal sensor-fusion pipeline for tracking, analyzing, and correcting strength-training biomechanics.
Main Hardware	Wearable 9-axis IMUs such as MPU9250 or BNO085, ESP32 microcontrollers, RGB or RGB-D cameras, haptic feedback motors, and optional edge GPU hardware.
Main Software	Python, C++ or Arduino-style embedded firmware, FreeRTOS, MediaPipe, OpenPose, PyTorch, Extended Kalman Filter, Dynamic Time Warping, and optional mobile or web dashboard software.
Target Outputs	Real-time 3D joint-angle estimation, repetition counting, form-deviation scoring, fatigue warning, localized haptic feedback, and exercise-session analytics.
Faculty Review Focus	Mathematical rigor of sensor fusion, quaternion mathematics, latency optimization, validation against optical ground truth, magnetic interference resilience, anthropometric calibration, usability, and VIP continuity.
Evaluator	Prof. Mehdi Pirahandeh.
Total Grade	50 points.

3 Professor Mehdi Grading Summary

Professor Mehdi Grading Evaluation Summary

The following table converts the evaluation into a clear 50-point grading structure. The score reflects the team’s strong grasp of multisensor fusion, biomechanical theory, and applied AI, while highlighting the need for more rigorous environmental testing, especially magnetic interference in gym environments, strap slippage, latency measurement, and robust anthropometric calibration routines.

4 Table of Notations

5 System Architecture Overview

5.1 Hardware and Edge Data Acquisition Infrastructure

The physical infrastructure of Team “The Last Set” is bifurcated into a localized wearable sensor array and a stationary edge-vision node. This hybrid approach is critical for resolving the inherent limitations of each respective technology. Wearable IMUs provide high-frequency data that is excellent for capturing rapid motion, but they suffer from integration drift and magnetic disturbance. Optical pose estimation provides global spatial reference, but it is vulnerable to occlusion and camera-angle limitations.

Architecture Note Hardware Layer

Wearable IMUs suffer from integration drift over time, particularly on the yaw axis. Conversely, optical computer vision suffers from occlusion when limbs cross the body or are blocked by gym equipment. Fusing these two data streams provides absolute spatial referencing combined with high-frequency kinematic resolution.

- **Wearable Sensor Nodes:** Custom-designed printed circuit boards housing an ESP32 microcontroller, an MPU9250 or BNO085 9-axis IMU, a small battery, and a localized haptic motor. These nodes communicate through low-latency Bluetooth Low Energy or UDP over Wi-Fi.
- **Stationary Edge Vision Node:** An NVIDIA Jetson Orin, high-end laptop, or comparable edge device equipped with a standard RGB or RGB-D camera. This node processes MediaPipe or OpenPose pose estimation at a

Table 2: Professor Mehdi Pirahandeh's Final Grading Summary

Evaluation Dimension	Max	Score	Professor Mehdi's Evaluation Summary
VIP Continuity and System Foundation	10	9.5	The modular separation of wearable hardware from the vision processing node provides an excellent foundation for future VIP cohorts to iterate independently on embedded sensing, optical pose estimation, and feedback interfaces.
Technical Rigor and Research Depth	10	9.0	The integration of EKF for quaternion estimation and DTW for temporal sequence alignment demonstrates a high level of mathematical maturity, signal-processing awareness, and biomechanical relevance.
Execution, Testing, and Practical Feasibility	10	8.0	The theoretical design is excellent, but the testing plan must account for ferromagnetic gym environments, strap slippage, long-session sensor drift, sweat exposure, and ground-truth validation.
Software Pipeline and Automation Readiness	8	8.0	The dual-threaded architecture, with real-time firmware on ESP32 and higher-level analysis on an edge node, is well conceived. However, BLE or Wi-Fi synchronization protocols require careful optimization.
Innovation, Commercial Potential, and Future Roadmap	7	6.5	The application to physical therapy, commercial fitness, and sports performance is highly promising. The roadmap toward personalized anthropometric scaling and predictive injury modeling is strong.
Documentation and Presentation Quality	5	3.5	The report establishes clear technical constraints, rigorous mathematical definitions, and a logical progression of engineering milestones.
Total	50	44.5	Outstanding interdisciplinary engineering work bridging embedded systems, AI, sensor fusion, and biomechanics.

Table 3: Symbols, Abbreviations, and Key Technical Terms

Symbol or Term	Definition
IMU	Inertial Measurement Unit; a sensor package consisting of an accelerometer, gyroscope, and often a magnetometer.
EKF	Extended Kalman Filter; a nonlinear state estimator used for sensor fusion, drift correction, and noise reduction.
DTW	Dynamic Time Warping; an algorithm for measuring similarity between two temporal sequences that may vary in speed.
DoF	Degrees of Freedom; the independent ways in which a mechanical or biomechanical system can move.
DSP	Digital Signal Processing; mathematical processing of sampled signals such as acceleration and angular velocity.
PPG	Photoplethysmography; optical measurement technique commonly used for heart-rate estimation.
BLE	Bluetooth Low Energy; a low-power wireless communication protocol suitable for wearable sensors.
RGB-D	Red, Green, Blue, and Depth camera; a camera system that captures both color and depth information.
$\mathbf{q}$	Quaternion representing 3D spatial orientation, usually defined as $[q_w, q_x, q_y, q_z]^T$.
$\boldsymbol{\omega}$	Angular velocity vector measured by the gyroscope, $[\omega_x, \omega_y, \omega_z]^T$.
$\mathbf{a}$	Linear acceleration vector measured by the accelerometer.
$\mathbf{m}$	Magnetic field vector measured by the magnetometer.
T_i^{i-1}	Denavit–Hartenberg transformation matrix between adjacent links in a kinematic chain.
RMSE	Root Mean Square Error; a metric for measuring average estimation error compared to ground truth.

minimum target rate of 30 frames per second.

- **Central Processing Hub:** The edge node acts as the central coordinator, aligning high-frequency IMU data sampled around 100 Hz with lower-frequency optical data around 30 Hz using timestamp synchronization, complementary filtering, or an EKF.
- **Feedback Interface:** The system should support haptic vibration, audio alerts, or visual dashboard feedback. Real-time feedback should be localized and immediate because delayed feedback after a repetition is much less useful.
- **Optional Mobile Dashboard:** A mobile or web interface may provide repetition history, joint-angle plots, form-score trends, and exercise-specific coaching messages.

5.2 System Layering and Data Responsibilities

Table 4: Layered System Architecture

Layer	Main Components	Engineering Responsibility
Wearable Sensing Layer	ESP32, IMU, battery, haptic motor	Samples acceleration, angular velocity, magnetic field, and sends timestamped packets.
Vision Layer	RGB or RGB-D camera, MediaPipe, OpenPose	Estimates body landmarks and provides absolute spatial reference for drift correction.
Fusion Layer	EKF, complementary filter, timestamp alignment	Combines IMU and vision data into stable joint orientation estimates.
Biomechanics Layer	Forward kinematics, joint-angle calculation, anthropometric scaling	Converts orientation estimates into interpretable biomechanical variables.
Scoring Layer	DTW, form score, repetition segmentation	Compares the user's motion with golden reference repetitions and quantifies deviation.
Feedback Layer	Haptic motor, audio cue, dashboard	Provides immediate corrective feedback to reduce injury risk and improve form.

5.3 End-to-End Kinematic Processing Flowchart

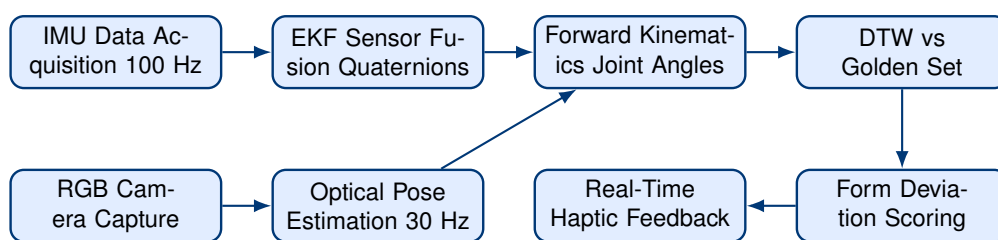


Figure 1: End-to-end data processing pipeline fusing high-frequency inertial data with optical tracking to evaluate biomechanical form.

6 Algorithms and Mathematical Foundations

6.1 Quaternion Mathematics and Kinematic Orientation

To avoid the singularity problem inherent in Euler angles, commonly known as gimbal lock, the system should use quaternions to represent the 3D orientation of human limbs. A quaternion $\mathbf{q}$ is defined as:

$$\mathbf{q} = q_w + q_x \mathbf{i} + q_y \mathbf{j} + q_z \mathbf{k}. \quad (1)$$

The temporal evolution of the quaternion based on the angular velocity ω measured by the gyroscope is governed by:

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \mathbf{p}(\omega), \quad (2)$$

where:

$$\mathbf{p}(\omega) = [0, \omega_x, \omega_y, \omega_z]^T \quad (3)$$

is the pure quaternion representation of angular velocity, and $\otimes$ denotes quaternion multiplication.

In discrete time, assuming constant angular velocity over a sampling interval Δt , the quaternion update rule can be expressed as:

$$\mathbf{q}_{t+1} = \mathbf{q}_t \otimes \begin{bmatrix} \cos\left(\frac{\|\omega\|\Delta t}{2}\right) \\ \frac{\omega_x}{\|\omega\|} \sin\left(\frac{\|\omega\|\Delta t}{2}\right) \\ \frac{\omega_y}{\|\omega\|} \sin\left(\frac{\|\omega\|\Delta t}{2}\right) \\ \frac{\omega_z}{\|\omega\|} \sin\left(\frac{\|\omega\|\Delta t}{2}\right) \end{bmatrix}. \quad (4)$$

Mathematical Interpretation Quaternion Integration

Pure integration of gyroscope data accumulates error over time. To correct this drift, the system should use the accelerometer as a gravity reference and the magnetometer as a heading reference. However, in gyms, magnetic readings can be unreliable because iron plates and steel racks distort the magnetic field. This makes optical correction particularly important.

6.2 Extended Kalman Filter Formulation

The state vector $\mathbf{x}$ consists of the orientation quaternion and gyroscope biases:

$$\mathbf{x} = [q_w, q_x, q_y, q_z, b_x, b_y, b_z]^T. \quad (5)$$

The nonlinear state transition model $f(\mathbf{x}, \mathbf{u})$ predicts the next state based on the gyroscope input $\mathbf{u}_k = \omega_k$:

$$\mathbf{x}_{k|k-1} = f(\mathbf{x}_{k-1|k-1}, \mathbf{u}_k). \quad (6)$$

The predicted covariance matrix $\mathbf{P}$ is updated using the Jacobian $\mathbf{F}_k$ of the transition function:

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^T + \mathbf{Q}, \quad (7)$$

where $\mathbf{Q}$ is the process noise covariance matrix.

The measurement update compares the predicted gravity and magnetic vectors against the actual readings from the accelerometer $\mathbf{a}$ and magnetometer $\mathbf{m}$. The Kalman gain $\mathbf{K}_k$ minimizes the posterior error covariance:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R})^{-1}, \quad (8)$$

$$\mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - h(\mathbf{x}_{k|k-1})), \quad (9)$$

where $\mathbf{H}_k$ is the Jacobian of the measurement function $h(\cdot)$, and $\mathbf{R}$ is the measurement noise covariance.

Technical Recommendation EKF Robustness

The EKF should include a magnetic-anomaly rejection mechanism. If the magnetometer magnitude or direction deviates sharply from expected Earth-field values, the filter should reduce magnetometer weight and rely more strongly on gyroscope integration, accelerometer gravity reference, and optical pose correction.

6.3 Forward Kinematics for Human Movement

After estimating segment orientations, the system must translate sensor-level orientation data into meaningful biomechanical joint angles. A simplified kinematic chain can represent the human body as linked rigid segments. For adjacent links, the homogeneous transformation matrix is commonly expressed as:

$$T_i^{i-1} = \begin{bmatrix} \mathbf{R}_i^{i-1} & \mathbf{d}_i^{i-1} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad (10)$$

where $\mathbf{R}_i^{i-1}$ is the relative rotation matrix between neighboring segments, and $\mathbf{d}_i^{i-1}$ is the translation vector representing limb length or joint offset.

The complete chain transformation from the body root to segment n is:

$$T_n^0 = T_1^0 T_2^1 \cdots T_n^{n-1}. \quad (11)$$

Mathematical Interpretation Biomechanical Interpretation

Forward kinematics converts sensor readings into meaningful exercise metrics, such as knee flexion, hip extension, trunk inclination, shoulder abduction, and elbow angle. These values are far more useful for coaching than raw accelerometer readings.

6.4 Sequence Alignment via Dynamic Time Warping

A primary challenge in biomechanical analysis is that users perform repetitions at different speeds. Comparing a user's 3-second squat to a professional's 2-second "golden" squat using strict Euclidean distance produces false-positive errors due to phase misalignment.

Let:

$$X = [x_1, x_2, \dots, x_N]$$

be the time series of the user's joint angles, and:

$$Y = [y_1, y_2, \dots, y_M]$$

be the golden reference sequence. DTW constructs an $N \times M$ cost matrix D , where each element is defined recursively:

$$D(i, j) = c(x_i, y_j) + \min \begin{cases} D(i-1, j) & \text{insertion,} \\ D(i, j-1) & \text{deletion,} \\ D(i-1, j-1) & \text{match.} \end{cases} \quad (12)$$

Here:

$$c(x_i, y_j) = \|x_i - y_j\|_2 \quad (13)$$

is the local Euclidean distance between joint-angle vectors at respective frames. The normalized distance of the warping path provides a robust form score independent of repetition tempo.

7 Reference Pseudocode and Core Workflows

7.1 Python Formulation for DTW Form Scoring

The following Python script illustrates how the team can implement the DTW algorithm to evaluate a stream of incoming kinematic joint angles against a pre-recorded standard.

```
1 import numpy as np
2
```



```

3
4 def compute_dtw_distance(user_seq: np.ndarray, golden_seq: np.ndarray) -> float:
5     """
6     Compute the Dynamic Time Warping distance between two kinematic sequences.
7
8     Both inputs should be arrays of shape (time_steps, features), where
9     each feature can represent a joint angle or biomechanical variable.
10    """
11    if user_seq.ndim != 2 or golden_seq.ndim != 2:
12        raise ValueError("Both sequences must be 2D arrays.")
13
14    if user_seq.shape[1] != golden_seq.shape[1]:
15        raise ValueError("Both sequences must have the same feature dimension.")
16
17    n_steps = len(user_seq)
18    m_steps = len(golden_seq)
19
20    dtw_matrix = np.full((n_steps + 1, m_steps + 1), np.inf)
21    dtw_matrix[0, 0] = 0.0
22
23    for i in range(1, n_steps + 1):
24        for j in range(1, m_steps + 1):
25            cost = np.linalg.norm(user_seq[i - 1] - golden_seq[j - 1])
26
27            dtw_matrix[i, j] = cost + min(
28                dtw_matrix[i - 1, j],
29                dtw_matrix[i, j - 1],
30                dtw_matrix[i - 1, j - 1],
31            )
32
33    normalized_distance = dtw_matrix[n_steps, m_steps] / (n_steps + m_steps)
34    return float(normalized_distance)
35
36
37 def evaluate_repetition(
38     user_data: np.ndarray,
39     golden_data: np.ndarray,
40     threshold: float = 15.0
41 ) -> bool:
42     """
43     Evaluate one repetition and trigger feedback when form degrades.
44
45     A lower score indicates stronger similarity to the golden reference.
46     """
47     score = compute_dtw_distance(user_data, golden_data)
48     print(f"Repetition Form Score: {score:.2f}")
49
50     if score > threshold:
51         print("WARNING: Form degradation detected. Triggering haptic feedback.")
52         # trigger_esp32_haptic_motor()
53         return False
54
55     print("Form accepted.")
56     return True
57
58
59 if __name__ == "__main__":
60     user_sequence = np.array([
61         [10, 20, 30],
62         [15, 25, 35],
63         [20, 30, 40],

```

```

64     ], dtype=float)
65
66     golden_sequence = np.array([
67         [11, 19, 29],
68         [16, 24, 36],
69         [21, 31, 39],
70     ], dtype=float)
71
72     evaluate_repetition(user_sequence, golden_sequence)

```

Listing 1: Dynamic Time Warping for Biomechanical Form Scoring

7.2 ESP32 Data Acquisition Loop

The firmware on the ESP32 must operate with strict timing to ensure valid numerical integration. Inconsistent sampling intervals can corrupt quaternion integration and lead to unstable joint-angle estimates.

```

1  #include <Wire.h>
2  #include <MPU9250.h>
3
4  MPU9250 imu;
5
6  unsigned long lastTime = 0;
7  const int SAMPLE_RATE_HZ = 100;
8  const int SAMPLE_INTERVAL_MS = 1000 / SAMPLE_RATE_HZ;
9
10 void setup() {
11     Serial.begin(115200);
12     Wire.begin();
13
14     if (!imu.setup(0x68)) {
15         Serial.println("IMU_initialization_failed.");
16         while (1) {
17             delay(1000);
18         }
19     }
20
21     lastTime = millis();
22 }
23
24 void loop() {
25     unsigned long currentTime = millis();
26
27     if (currentTime - lastTime >= SAMPLE_INTERVAL_MS) {
28         lastTime = currentTime;
29
30         if (imu.update()) {
31             Serial.print(imu.getAccX()); Serial.print(",");
32             Serial.print(imu.getAccY()); Serial.print(",");
33             Serial.print(imu.getAccZ()); Serial.print(",");
34             Serial.print(imu.getGyroX()); Serial.print(",");
35             Serial.print(imu.getGyroY()); Serial.print(",");
36             Serial.print(imu.getGyroZ()); Serial.print(",");
37             Serial.print(imu.getMagX()); Serial.print(",");
38             Serial.print(imu.getMagY()); Serial.print(",");
39             Serial.println(imu.getMagZ());
40
41             // In full deployment, send the same data via UDP or BLE
42             // to the local edge-processing node.
43         }

```

```
44     }  
45 }
```

Listing 2: Embedded C/C++ Pseudo-Firmware for High-Frequency IMU Sampling

8 Testing and Quality Assurance

8.1 Validation Against Optical Ground Truth

To validate the accuracy of IMU-derived kinematics, the team should employ an industry-standard optical motion capture system, such as Vicon or OptiTrack, as the ground truth. If such a system is unavailable, the team can begin with high-frame-rate camera recordings and manually labeled joint angles, but a professional validation path should eventually include optical motion capture.

Risk and Limitation Magnetic Interference in Gym Environments

Commercial gyms contain large masses of moving ferromagnetic materials, including iron plates, steel racks, cable machines, and barbells. These moving masses continuously distort the local magnetic field, invalidating the magnetometer’s assumption of a static magnetic North. The EKF must dynamically reject anomalous magnetic readings or rely more heavily on optical vision for yaw correction when the sensor is near iron equipment.

Accuracy Metrics: Testing protocols should measure the Root Mean Square Error (RMSE) of critical joint angles, including knee flexion, hip extension, lumbar inclination, shoulder abduction, and elbow flexion across at least 500 repetitions of various exercises. An acceptable engineering tolerance for a prototype system is an RMSE below 3.5° compared to optical ground truth.

8.2 Testing and Quality Assurance Plan

Table 5: Testing and Quality Assurance Plan

Test Area	Method	Acceptance Criterion
Sensor Fusion RMSE	Compare EKF output against Vicon, OptiTrack, or carefully labeled optical tracking data.	Joint-angle error should remain $\leq 3.5^\circ$ over a 60-second trial.
Latency Analysis	Measure time from physical movement to haptic trigger.	Total sensing, processing, and feedback latency should remain ≤ 50 ms.
Magnetic Rejection	Introduce a 20 kg iron plate near the sensor during operation.	Algorithm should reject magnetic disturbance without catastrophic yaw divergence.
DTW Robustness	Perform repetitions at 0.5×, 1.0×, and 2.0× normal speed.	Form-score variance should remain below 10 percent for equivalent movement quality.
Battery Endurance	Perform continuous sampling and BLE or Wi-Fi transmission testing.	ESP32 wearable nodes should operate for more than 4 hours continuously.
Strap Slippage	Deliberately shift the sensor strap during exercise trials.	System should detect discontinuous orientation jumps and warn the user.
Sweat and Motion Stress	Test with repeated high-motion sessions and sweat-like moisture exposure.	Enclosure and sensor output should remain stable and electrically safe.

8.3 Recommended Performance Metrics

Table 6: Recommended Quantitative Performance Metrics

Metric	Purpose	Recommended Reporting Method
Joint-Angle RMSE	Measures accuracy of estimated angles.	Report per joint and per exercise.
Feedback Latency	Measures usefulness of real-time correction.	Report median, 95th percentile, and worst-case latency.
Repetition Classification Accuracy	Measures ability to identify correct and incorrect form.	Use labeled repetitions with confusion matrix, precision, recall, and F1-score.
Drift Rate	Measures orientation degradation over time.	Report yaw, pitch, and roll drift per minute.
Sensor Packet Loss	Measures wireless reliability.	Report packet-loss rate during exercise and under Wi-Fi/BLE congestion.
User Comfort Score	Measures wearable usability.	Collect survey ratings for strap comfort, weight, and feedback clarity.

9 Technical Recommendations and Future Work

The following recommendations establish a rigorous multi-semester technical roadmap designed to mature the project from a prototype into a clinically and commercially viable product.

9.1 Phase 1: Anthropometric Scaling and Initialization

Currently, kinematic calculations may assume standard limb lengths. To accurately model torque and joint load, future cohorts should implement automated anthropometric scaling. Using the initial frames from the optical camera, the system should estimate user height and derive segment lengths such as femur, tibia, humerus, forearm, and torso based on established biomechanical tables, including Dempster-style body segment parameters.

Technical Recommendation Biomechanical Load Estimation

By scaling the kinematic model with the user's estimated limb lengths and inputting the external weight being lifted, the system can calculate dynamic joint torques using inverse dynamics. This allows the software to warn users if compressive or shear load on the lumbar spine, especially around L4/L5, exceeds safe physiological limits.

9.2 Phase 2: Transition to Federated Learning

As the dataset of user repetitions grows, relying on a static golden set becomes rigid. Expert execution varies based on body proportions. For example, a tall lifter may have a different optimal deadlift trajectory than a short lifter. The team should implement clustering algorithms such as K-Means or DBSCAN on the cloud server to categorize golden trajectories by user somatotype, exercise type, training experience, and mobility constraints.

Furthermore, Federated Learning would allow edge nodes to update their local classification models without transmitting raw privacy-sensitive video data to the cloud. This is especially valuable for clinical rehabilitation, where patient movement data may be sensitive.

9.3 Phase 3: Deep Neural Networks for Predictive Injury Modeling

Beyond threshold-based form checking, future VIP students should construct LSTM networks or Temporal Convolutional Networks trained on sequential kinematic data. These models can identify micro-degradations in form over the course of a workout due to muscular fatigue, preemptively warning the user before catastrophic technique breakdown or injury occurs.

9.4 Phase 4: Clinical Rehabilitation Dashboard

A future clinical version should provide therapists with range-of-motion summaries, adherence tracking, pain-report integration, session-to-session progress, and automated abnormality flags. This would shift the system from a fitness-only product toward a regulated rehabilitation-support tool.

Technical Recommendation Clinical Expansion

For rehabilitation use, the system must emphasize explainability, repeatability, and conservative safety thresholds. Clinicians need interpretable joint-angle graphs, not only an AI-generated form score.

10 Business and Market Analysis

Business and Market Insight

The commercial feasibility of a highly accurate, AI-driven biomechanical tracking system is substantial. Current market offerings rely either on expensive camera setups, such as smart home gyms, or basic IMU wearables that lack spatial awareness. Team “The Last Set” bridges this gap by proposing an affordable hybrid architecture combining inertial sensing and computer vision.

The project’s architecture caters to two distinct B2B and B2C markets:

1. **Clinical Rehabilitation (B2B):** Physical therapists can use the system to assign specific movement protocols. The system objectively logs range-of-motion progression over weeks, allowing data-driven recovery tracking.
2. **Commercial Fitness (B2C):** As a SaaS application integrated with wearable sensors, users receive coaching and injury-prevention logic previously reserved for professional athletes with access to biomechanics laboratories.
3. **Sports Performance Labs (B2B):** Coaches can use the system to analyze asymmetry, fatigue, and technical drift during repeated training sessions.
4. **Workplace Ergonomics (B2B):** Industrial safety teams can monitor repetitive lifting patterns and reduce musculoskeletal injury risk.

To become commercially viable, the project should prioritize comfort, low setup friction, fast calibration, interpretable feedback, and privacy-preserving data storage. Users will not adopt a system that requires long calibration before every workout or produces confusing feedback.

11 Risk and Mitigation Analysis

12 Implementation Roadmap for Future VIP Cohorts

13 Final Professor Note

Executive Summary

Team “The Last Set” presents an outstanding interdisciplinary VIP project that combines embedded systems, wearable sensing, computer vision, signal processing, biomechanics, and applied AI. The strongest aspect of the proposal is the multimodal architecture, which correctly recognizes that neither IMU-only nor vision-only systems are sufficient for reliable strength-training analysis. To mature into a clinically and commercially viable system, the project must focus on magnetic interference rejection, optical ground-

Table 7: Major Technical Risks and Mitigation Strategies

Risk	Impact	Mitigation
Quaternion Drift	Accumulated integration error renders joint angles invalid after several minutes.	Use absolute orientation resets from optical vision, robust EKF tuning, and periodic stationary recalibration.
Vision Occlusion	Camera loses sight of joints when the user squats inside a power rack or limbs cross the body.	Rely more heavily on IMU dead reckoning during low optical confidence and restore correction when landmarks reappear.
Magnetic Interference	Iron plates and steel gym equipment corrupt magnetometer heading.	Implement magnetic anomaly detection and reduce magnetometer weight during disturbed intervals.
Sweat Ingress	Moisture can short-circuit ESP32 or IMU wearable nodes.	Use sealed enclosures, conformal coating, epoxy potting, or IP-rated wearable casing.
High Latency	Haptic feedback arrives after the repetition is finished, reducing usefulness.	Execute EKF and DTW locally on the edge node and avoid cloud processing in the real-time loop.
Strap Slippage	Sensor physically rotates on the user's limb during exercise.	Use high-friction straps and detect sudden discontinuous jumps in orientation.
Poor Calibration	Incorrect initial posture or limb-length assumptions produce invalid form scores.	Add guided calibration, anthropometric scaling, and calibration-quality checks.
User Overdependence	Users may rely on the system instead of professional medical guidance.	Clearly position the system as coaching support, not medical diagnosis, unless clinically validated.

Table 8: Multi-Phase VIP Implementation Roadmap

Phase	Primary Goal	Expected Deliverables
1	Wearable Node Hardware and Baseline Data Acquisition	Functioning ESP32 and IMU prototype, reliable BLE or Wi-Fi streaming, raw data visualization dashboard, and battery endurance report.
2	Sensor Fusion and Kinematics	Implementation of EKF, stable quaternion orientation estimation, and transformation of raw sensor data into interpretable joint-angle values.
3	Vision Integration and DTW Scoring	Fusion of MediaPipe or OpenPose data with IMU data, functioning DTW algorithm, real-time form score, and haptic alert prototype.
4	Inverse Dynamics and Joint Torques	Anthropometric scaling, dynamic joint-load estimation, lumbar risk warnings, and exercise-specific biomechanical reports.
5	Predictive AI and Clinical Dashboards	LSTM or TCN fatigue prediction, cloud-based therapist portal, patient progress dashboard, and privacy-preserving analytics pipeline.
6	Commercial Pilot and Human-Factors Testing	User comfort testing, calibration usability study, gym-environment robustness validation, and minimum viable product packaging.

truth validation, strap-slippage detection, latency measurement, anthropometric calibration, and explainable feedback. With these refinements, this project can become a flagship VIP platform for digital fitness, rehabilitation technology, and AI-driven biomechanics.